

Relativita

$$x \rightarrow t, \vec{x} \quad \phi = \frac{q}{4\pi\epsilon_0} \frac{1}{r(1 - \frac{\vec{v} \cdot \vec{v}}{c^2})} \Big|_{t_{ret}}$$

$$\vec{A} = \frac{q}{4\pi\epsilon_0} \frac{\vec{v}}{r(1 - \frac{\vec{v} \cdot \vec{v}}{c^2})} \Big|_{t_{ret}}$$

$$\vec{E} = -\vec{\nabla}\phi - \frac{\partial\vec{A}}{\partial t}$$

$$\vec{B} = \vec{\nabla} \times \vec{A}$$

$$t_{ret} = t - \frac{r_{ret}}{c} \quad (*)$$

$$c(t - t_{ret}) = r_{ret} = |\vec{x} - \vec{x}_E(t_{ret})|$$

$$\frac{1}{c} \frac{\partial}{\partial t} (*) = 1 - \frac{\partial t_{ret}}{\partial t} = \frac{1}{c} \frac{1}{r_{ret}} \vec{r}_{ret} \cdot \frac{\partial \vec{r}_{ret}}{\partial t}$$

$$= \frac{1}{c} \vec{e}_{ret} \cdot \vec{v}(t_{ret}) \frac{\partial t_{ret}}{\partial t}$$

$$\Rightarrow \frac{\partial t_{ret}}{\partial t} = \frac{1}{1 - \frac{\vec{e} \cdot \vec{v}}{c}} \Big|_{t_{ret}}$$

$$\vec{\nabla} (*) = -\vec{\nabla}(ct_{ret}) = \frac{1}{r_{ret}} (\vec{\nabla}(\vec{x} - \vec{x}_E(t_{ret})) \cdot \vec{r}_{ret})$$

$$= (\vec{I} - (\vec{\nabla} t_{ret}) \vec{v}) \cdot \vec{e}_{ret}$$

$$= \vec{e}_{ret} - (\vec{\nabla} t_{ret}) \vec{v} \cdot \vec{e} \Big|_{ret}$$

$$\Rightarrow \vec{\nabla} t_{ret} = - \frac{\vec{e}}{1 - \frac{\vec{e} \cdot \vec{v}}{c}} \Big|_{t_{ret}}$$

$$\Rightarrow \vec{\nabla} r_{ret} = \frac{\vec{e}}{1 - \frac{\vec{e} \cdot \vec{v}}{c}} \Big|_{t_{ret}}$$

$$\frac{\partial r_{ret}}{\partial t} = - \frac{\vec{e} \cdot \vec{v}}{1 - \frac{\vec{e} \cdot \vec{v}}{c}} \Big|_{t_{ret}}$$

$$\Rightarrow \vec{E} = \frac{q}{4\pi\epsilon_0} \frac{1}{(1 - \frac{\vec{e} \cdot \vec{v}}{c})^3} \left[\frac{\vec{e} - \frac{\vec{v}}{c}}{r^2} + \frac{\vec{e} \times ((\vec{e} - \frac{\vec{v}}{c}) \times \vec{v})}{c^2 r} \right]_{t_{ret}}$$

$$\vec{B} = \frac{1}{c} \vec{e}_{ret} \times \vec{E}$$

Distinze k retardovaním a odhazováním řešení

$$A^M = A_{ret}^M + A_{in}^M$$

$$A_{adv}^M = A_{ret}^M - A_{in}^M$$

$$-\square A_{ret}^M = \frac{1}{\epsilon_0 c^2} j^M$$

$$\square A_{in}^M = 0$$

$$\square A_{adv}^M = \square A_{ret}^M - \square A_{in}^M = \frac{1}{\epsilon_0 c^2} j^M - 0 = \frac{1}{\epsilon_0 c^2} j^M$$

$$A_{out} = A_{ret} - A_{adv} + A_{in} = A_{ret} - (A_{ret} - A_{in}) + A_{in} = A_{in} + A_{in} = 2A_{in}$$

→ uvažuj se nos klesání
 ⇒ předpokládá se na začátku
 to bylo buď předchozí
 nebo počáteční pole nejsou
 klesající s budoucí
 zloží

$$A_{adv}(x) = \frac{1}{\epsilon_0 c^2} \int G_{adv}(x|x') j(x') d^4\Omega'$$

$$A_{ret}(x) = \frac{1}{\epsilon_0 c^2} \int (G_{ret}(x|x') - G_{adv}(x|x')) j(x') d^4\Omega'$$

$$\square G_c(x|x') = 0$$

$$G_c(x|x') = -G_c(x'|x)$$

$$\square A^M = 0 \quad \text{dává na } \Sigma \quad A^M(x) \quad \frac{\partial A^M}{\partial n}(x)$$

$$\int_{\Omega} (\phi \square \eta - \eta \square \phi) d^4\Omega =$$

$$= - \int_{\Sigma_k} (\phi \frac{\partial \eta}{\partial n} - \eta \frac{\partial \phi}{\partial n}) d^3V + \int_{\Sigma_z} (\phi \frac{\partial \eta}{\partial n} - \eta \frac{\partial \phi}{\partial n}) d^3V$$

$$1) \quad x \in \langle \Sigma_k | \Sigma \rangle \quad \eta(x) \leftrightarrow A^M(x)$$

$$\phi(x') = G_{ret}(x|x')$$

$$A^M(x) = \int_{\langle \Sigma_k | \Sigma \rangle} \left[G_{ret}(x|x') \underbrace{\square' A^M(x')}_0 - A^M(x') \underbrace{\square' G_{ret}(x|x')}_{-\delta(x|x')} \right] d^4\Omega'$$

$$= \int_{x' \in \Sigma} \left[G_{ret}(x|x') \frac{\partial A^M}{\partial n'}(x') - A^M(x') \frac{\partial G_{ret}(x|x')}{\partial n'} \right] dV$$

$$- \int_{x' \in \Sigma_k} \left[\underbrace{G_{ret}}_0 \underbrace{\frac{\partial A^M}{\partial n'}}_{\text{konstanta struktura}} - A^M(x') \underbrace{\frac{\partial G_{ret}}{\partial n'}}_0 \right] dV$$

$$\Rightarrow \int_{x' \in \Sigma} \left[G_{ret}(x|x') \frac{\partial A^M}{\partial n'}(x') + \left(\frac{\partial G_{ret}}{\partial n'} \right)(x|x') A^M(x') \right] dV'$$

$$= \begin{cases} A^M(x) & \times \text{ po } \Sigma \\ 0 & \times \text{ před } \Sigma \end{cases}$$

$$\int_{x' \in \Sigma} \left[G_{adv}(x|x') \frac{\partial A^M}{\partial n'}(x') + \left(\frac{\partial G_{adv}}{\partial n'} \right)(x|x') A^M(x') \right] dV'$$

$$= \begin{cases} 0 & \times \text{ po } \Sigma \\ A^M(x) & \times \text{ před } \Sigma \end{cases}$$

$$\Rightarrow A^M(x) = \int_{x' \in \Sigma} \left[G_c(x|x') \frac{\partial A^M}{\partial n'}(x') + \left(\frac{\partial G_c}{\partial n'} \right)(x|x') A^M(x') \right] dV'$$

$$x, x' \in \Sigma$$

Brzdící síla

$$F = F_{ret} + F_{in} \quad \text{pole kolem náboje}$$

$$f = qF \cdot u = qF_{ret} \cdot u + qF_{in} \cdot u$$

→ symetrický příspěvek už nezapínej! ⇒ vyhodíme je,
 zapínej! už s pouze nevyhodíme spotřebu s polynomem,
 ta působí na náboj

$$F_{ret} = F_{sym} + \frac{1}{2} F_{rad}$$

$$f_{self} = \frac{q}{2} F_{rad} \cdot u = \frac{q^2}{4\pi\epsilon_0} \left[\frac{1}{2} \frac{1}{3} \ddot{x}_E + \frac{2}{3} (\ddot{x}_E - \dot{x}_E^2 \dot{x}_E) + O(3) \right]$$

$$M \cdot \ddot{x}_E = f_{self} + f_{ext}$$

$$\Rightarrow m_0 \ddot{x}_E = f_{self} + f_{ext}$$

$$f_{self} = \frac{q^2}{4\pi\epsilon_0} \left[\frac{2}{3} (\ddot{x}_E - \dot{x}_E^2 \dot{x}_E) + O(3) \right]$$

$$m_0 = M_0 + \frac{1}{2} \frac{1}{3} \frac{q^2}{4\pi\epsilon_0}$$

$$\Rightarrow m_0 a = f_{ext} + \frac{2}{3} \frac{q^2}{4\pi\epsilon_0 m_0} m_0 (\ddot{a} - \dot{a}^2 \dot{a})$$

$$= f_{ext} + \tau_0 (\ddot{a} + \dot{a} \dot{a}) \cdot (m_0 a)$$

$$u^2 = -1$$

$$u \cdot a = 0$$

$$u \cdot \dot{a} + a^2 = 0$$

$$\tau_0 \sim 10^{-23} \text{ s}$$

$$= f_{ext} + \tau_0 (\ddot{a} + \dot{a} \dot{a}) \cdot (m_0 a)$$

$$\tau_0 \sim 10^{-23} \text{ s}$$

$$= f_{ext} + \tau_0 (\ddot{a} + \dot{a} \dot{a}) \cdot (m_0 a)$$

$$\tau_0 \sim 10^{-23} \text{ s}$$

$$= f_{ext} + \tau_0 (\ddot{a} + \dot{a} \dot{a}) \cdot (m_0 a)$$

$$\tau_0 \sim 10^{-23} \text{ s}$$

$$= f_{ext} + \tau_0 (\ddot{a} + \dot{a} \dot{a}) \cdot (m_0 a)$$

$$\tau_0 \sim 10^{-23} \text{ s}$$

$$= f_{ext} + \tau_0 (\ddot{a} + \dot{a} \dot{a}) \cdot (m_0 a)$$